9 Fig. 9.1 shows a mobile phone (cell phone) being charged on a wireless charging plate.

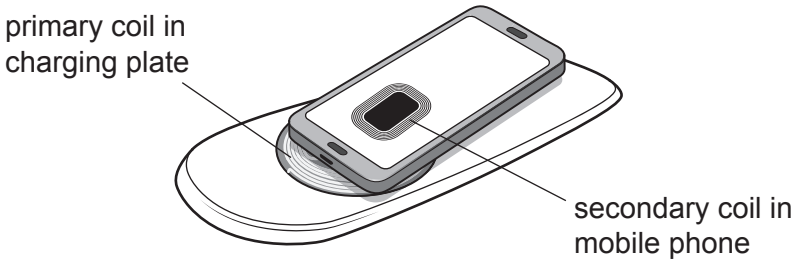


Fig. 9.1

(a) When the charging plate is switched on, there is an alternating current (a.c.) in the primary coil. A secondary coil is in the mobile phone.

Explain how a current is produced in the secondary coil.

.....

.....

..... [3]

(b) The maximum energy stored in the battery of the mobile phone is 0.012 kW h.

(i) Show that this maximum energy is 4.3×10^4 J.

[1]

(ii) The charging plate in Fig. 9.1 has a useful output power of 15 W.
The phone manufacturer claims that the battery can be charged to 50% capacity in less than 30 minutes.

Show that this claim is true.

[3]

[Total: 7]